Reinforcement learning algorithms solve Markov decision processes by estimating value functions or optimizing policies from sampled transitions. Two assumptions underlie the standard formulation. First, the agent's action at time t is determined solely by the observed state s_t and the agent's policy $\pi(a \mid s_t)$; there is no unobserved variable (confounder) simultaneously influencing both the action and the reward or state transition.[1] Second, the observed state s_t is sufficient for prediction, so that conditioning on s_t renders future states independent of past history. Both assumptions are the Markov property restated in causal language. Both fail routinely in applied settings.

This chapter studies what happens when logged sequential decisions are observational rather than experimental. The central object is the value of a target policy under the interventional law induced by that policy, not merely under the observational law generated by past behavior. The literature now covers confounded MDPs, partial identification and sensitivity analysis, proximal and negative-control methods for POMDPs, instrumental variables, mediator-based identification, and dynamic double machine learning for off-policy evaluation (OPE). The survey articles by Deng et al. (2023) and da Costa Cunha et al. (2025) provide broader coverage of causal representation learning, counterfactual policy optimization, transportability, and fairness.

Table 1 summarizes the main identification strategies. The purpose is not to create a checklist of papers. It is to show economists that many causal RL questions are familiar econometric problems once the estimand is written as a policy value rather than as a static treatment effect.

0.1 From Partial Observability to Causal Structure

A partially observable MDP (POMDP) augments the standard MDP with an observation function. The POMDP is defined by the tuple $(\mathcal{S}, \mathcal{A}, \mathcal{O}, P, O, r, \gamma)$, where $\mathcal{O}$ is a finite observation space and O is the observation function.

$$O(o \mid s', a) = P(O_t = o \mid S_t = s', A_{t-1} = a). \tag{1}$$

[1] Throughout this chapter, I reserve "outcome" for its causal inference meaning and use the specific RL quantity (reward, state, return, value) elsewhere.

Table 1: Econometric taxonomy for causal reinforcement learning

RL problem	Econometric analogue	Identifying object
Observed state blocks all confounding	Backdoor adjustment, sequential unconfoundedness, g-formula	Interventional transition and reward from conditional models given observed controls
Exogenous variation shifts actions	Instrumental variables and conditional moment restrictions	Structural transition or reward function satisfying moments conditional on instruments
Latent state is not observed but proxies are available	Proximal causal inference and negative controls	Bridge functions that recover policy value from proxy distributions
Actions operate through observed intermediate variables	Front-door, mediation, and g-methods	Mediated transition law and decomposed direct or indirect dynamic effects
No credible point-identifying source exists	Rosenbaum and Tan sensitivity, partial identification	Bounds over policy values under restricted hidden-bias models
High-dimensional nuisance functions are needed	DML, orthogonal scores, efficient OPE	Neyman-orthogonal moments for value or dynamic treatment parameters

The agent does not observe s_t directly but instead receives $o_t \sim O(\cdot \mid s_t, a_{t-1})$ and maintains a belief state $b_t \in \Delta(\mathcal{S})$

$$b_t(s) = P(S_t = s \mid o_1, a_1, \ldots, o_t), \tag{2}$$

which is updated via Bayesian filtering at each step. The belief MDP, whose state space is $\Delta(\mathcal{S})$, is itself a fully observable continuous-state MDP, so standard value iteration applies in principle, though computation is intractable in general.

da Costa Cunha et al. (2025) organize sequential decision problems with hidden variables into a hierarchy: standard MDP (full observability, no confounding), POMDP (partial observability, no confounding), confounded MDP (full observability, confounding), and causal POMDP (both). The key distinction is epistemic versus identificational. In a POMDP, the hidden state is a modeling challenge: the agent acknowledges incomplete information and plans accordingly via the belief state, analogous to Kalman or Hamilton filtering in econometrics. In a confounded MDP, the hidden variable is an identification challenge: standard estimators silently produce biased results, analogous to endogeneity and omitted variable bias.

This distinction matters because the same observed sequence can support different econometric interpretations. A hidden patient severity state may be a state variable the physician partially observes, a confounder that drives treatment and prognosis, a source of invalid support for a target policy, or a latent variable for which laboratory histories provide negative controls. Causal RL is the study of which interpretation is credible enough to identify, bound, or estimate the value of a counterfactual policy.

0.2 The Confounded MDP

When unobserved confounders influence both the behavioral policy and the transitions or rewards, the MDP is confounded. This formalization, developed by Zhang and Bareinboim (2019), Zhang and Bareinboim (2020), and Kallus and Zhou (2020), provides the foundation for causal reasoning in sequential decision problems. The key tool is Pearl's do-operator. The interventional distribution $P(Y \mid \mathrm{do}(X = x))$ is the distribution that arises when X is set externally rather than observed passively, severing all incoming causal influences on X while leaving the remaining data-generating process intact.[2]

Definition D1 (Confounded MDP (Zhang and Bareinboim, 2020))**.** *A confounded MDP is a tuple $(\mathcal{S}, \mathcal{A}, \mathcal{U}, P, r, \gamma)$ where $\mathcal{S}$ is a finite state space, $\mathcal{A}$ is a finite action space, $\mathcal{U}$ is a space of unobserved confounders, $\gamma \in [0, 1)$ is a discount factor, and the dynamics are governed by structural equations*

$$U_t \sim P_U(\cdot \mid S_t), \tag{3}$$

$$A_t \sim \mu(\cdot \mid S_t, U_t), \tag{4}$$

$$R_t = f_R(S_t, A_t, U_t) + \epsilon_t, \tag{5}$$

$$S_{t+1} \sim f_S(\cdot \mid S_t, A_t, U_t). \tag{6}$$

The behavioral (logging) policy μ depends on the unobserved confounder U_t through Equation (4). An evaluation policy $\pi(a \mid s)$ depends only on the observed state.

Unlike the online algorithms discussed earlier in the survey, where behavior and target

[2] The do-operator is formalized within the structural causal model (SCM) framework of Pearl (2009). An SCM specifies endogenous variables $\mathbf{V}$, exogenous variables $\mathbf{U}$, structural equations $V_i = f_i(\mathrm{pa}(V_i), U_i)$, and a distribution $P(\mathbf{U})$. The intervention $\mathrm{do}(X = x)$ replaces the structural equation for X with a constant, producing the interventional distribution. See Pearl (2009) for the complete framework, including the causal hierarchy (association, intervention, counterfactual) and general identification theory.

policies were identical or related by a known exploration mechanism, here μ is an unknown function of unobserved variables.

Because μ depends on U_t, conditioning on $\{A_t = a\}$ carries information about the confounder, so the observational and interventional transitions diverge:

$$P(s' \mid s, a) \neq P(s' \mid s, \mathrm{do}(a)). \tag{7}$$

The Bellman equation for policy evaluation must use interventional, not observational, transition probabilities. Define the causal Bellman operator for a *target policy* π:

Definition D2 (Causal Bellman Operator (Zhang and Bareinboim, 2020))**.** *The causal Bellman operator $\mathcal{T}_c^\pi$ for policy π in a confounded MDP is*

$$(\mathcal{T}_c^\pi V)(s) = \sum_{a \in \mathcal{A}} \pi(a \mid s) \sum_{s' \in \mathcal{S}} P(s' \mid s, \mathrm{do}(a)) \Big[r(s, a) + \gamma V(s') \Big], \tag{8}$$

where $r(s, a) = \mathbb{E}[R_t \mid S_t = s, \mathrm{do}(A_t = a)]$ is the interventional expected reward.

Lemma L1 (Bias of Naive Off-Policy Evaluation (Kallus and Zhou, 2020))**.** *Let $\hat{V}^\pi_{naive}$ denote the value function obtained by solving the Bellman equation with observational transitions $P(s' \mid s, a)$, and let V^π denote the true value function under interventional transitions $P(s' \mid s, \mathrm{do}(a))$. In a confounded MDP where $P(s' \mid s, a) \neq P(s' \mid s, \mathrm{do}(a))$ for some (s, a, s'), the naive estimator is biased.*

$$\hat{V}^\pi_{naive}(s) \neq V^\pi(s). \tag{9}$$

The importance-sampling estimator is also biased because the propensity $\mu(a \mid s)$ is not the true behavioral propensity $\mu(a \mid s, u)$:

$$\hat{V}^\pi_{IS} = \frac{1}{N} \sum_{i=1}^{N} \prod_{t=0}^{T} \frac{\pi(a_t^{(i)} \mid s_t^{(i)})}{\mu(a_t^{(i)} \mid s_t^{(i)})} G^{(i)}, \tag{10}$$

$$G^{(i)} = \sum_{t=0}^{T} \gamma^t R_t^{(i)}. \tag{11}$$

Here $G^{(i)}$ is the discounted return of trajectory i and T is the trajectory length.[3]

[3] This is the sequential analogue of the omitted variable bias in linear regression. In the static case, regressing Y on X without controlling for a confounder U yields a biased coefficient. In the sequential case, the bias propagates through the Bellman recursion and can amplify over the horizon.

The backdoor criterion (Pearl, 2009) provides a path to identification. Zhang and Bareinboim (2020) apply it to the confounded MDP setting.

Theorem 1 (Backdoor Identification in Confounded MDPs (Pearl, 2009; Zhang and Bareinboim, 2020))**.** *Suppose a set of observed variables $\mathbf{Z}_t$ satisfies the backdoor criterion relative to (A_t, S_{t+1}) in the causal graph of the confounded MDP, meaning that $\mathbf{Z}_t$ blocks all backdoor paths from A_t to S_{t+1} and no element of $\mathbf{Z}_t$ is a descendant of A_t.*[4] *Then the interventional transition probability is identified:*

$$P(s' \mid s, \mathrm{do}(a)) = \sum_{\mathbf{z}} P(s' \mid s, a, \mathbf{z})\, P(\mathbf{z} \mid s). \tag{12}$$

Substituting Equation (12) into the causal Bellman operator (Equation 8) yields an identified, consistent estimator of V^π.

0.3 Backdoor-Adjusted Off-Policy Evaluation

Off-policy evaluation under confounding is an average treatment effect estimation problem in a dynamic setting (Bannon et al., 2020): μ is the treatment assignment mechanism, π the counterfactual regime, importance sampling corresponds to inverse probability weighting, and doubly robust OPE corresponds to the AIPW estimator of Robins et al. (1994).

Theorem 1 yields a concrete estimation procedure. Given logged data D_N collected under behavioral policy μ, with elements $(s_t, a_t, z_t, r_t, s_{t+1})$ and observed proxy z_t, the econometrician estimates $\hat{P}(s' \mid s, a, z)$ and $\hat{P}(z \mid s)$ from the logged data, computes

$$\hat{P}(s' \mid s, \mathrm{do}(a)) = \sum_{z} \hat{P}(s' \mid s, a, z)\, \hat{P}(z \mid s), \tag{13}$$

and solves the causal Bellman equation using the estimated interventional transitions to obtain $\hat{V}^\pi$. The doubly robust variant combines the fitted action-value function $\hat{Q}(s,a)$ with backdoor-adjusted propensities, achieving consistency if either $\hat{Q}$ or the propensity model is correctly specified.

This is the cleanest case because the analyst observes enough variables to repair

[4] A backdoor path from A_t to S_{t+1} is any path in the causal graph that begins with an arrow into A_t, that is, a non-causal path. In the confounded MDP, $A_t \leftarrow U_t \rightarrow S_{t+1}$ is a backdoor path: U_t causes both A_t and S_{t+1}, creating a spurious association. Blocking all such paths by conditioning on appropriate variables eliminates the confounding bias. See Pearl (2009), Chapter 3.

the state. The rest of the chapter asks what can still be learned when that repair is unavailable, only partially credible, or statistically high-dimensional.

0.4 Alternative Identification Strategies

When no backdoor variable is available, causal RL moves from adjustment to sensitivity analysis, proxies, instruments, mediators, or orthogonal scores. Figure 1 displays the causal graph for three point-identification strategies used in the simulation study.

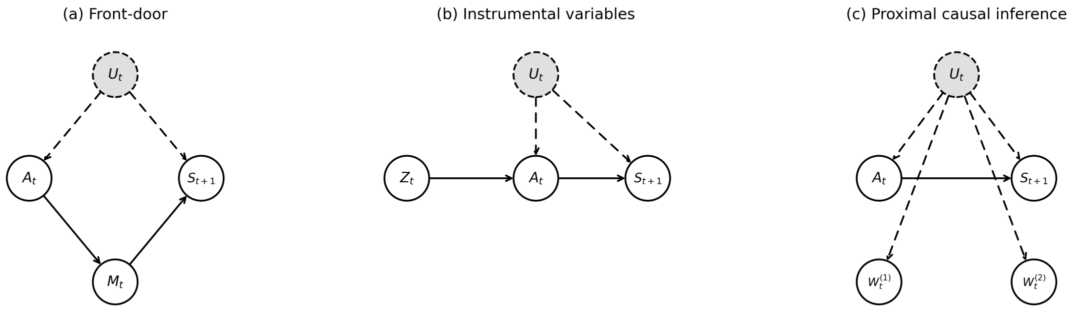

Figure 1: Causal graphs for three identification strategies in confounded MDPs. Gray dashed nodes are unobserved; dashed edges involve unobserved variables. (a) Front-door criterion with mediator M_t. (b) Instrumental variables with exogenous instrument Z_t. (c) Proximal causal inference with proxies $W_t^{(1)}, W_t^{(2)}$.

0.4.1 Sensitivity Analysis and Partial Identification

When point identification is not credible, the estimand becomes a set rather than a number. In static econometrics, this is the logic of Rosenbaum bounds, marginal sensitivity models, and partial identification (Rosenbaum, 2002; Tan, 2006; Manski, 2003). The sequential version asks how much a hidden variable could change the target policy value, allowing the hidden bias to enter each decision epoch. This subsection covers the three results that make this practical for off-policy evaluation in confounded MDPs.

Namkoong et al. (2020) provide the foundational result for finite-horizon sequential OPE under a Rosenbaum-style sensitivity model. Their identification assumption bounds the odds-ratio influence of any unobserved confounder by a parameter $\Gamma \geq 1$ at each decision epoch:

$$\Gamma^{-1} \;\leq\; \frac{\pi_t(a_t \mid H_t, u)/\pi_t(a_t' \mid H_t, u)}{\pi_t(a_t \mid H_t, u')/\pi_t(a_t' \mid H_t, u')} \;\leq\; \Gamma. \tag{14}$$

Under this Γ-bound, their Theorem 2 derives a lower bound on the target policy value $\mathbb{E}[Y(\bar{A}_{1:T})]$ as the solution to a tractable loss-minimization problem with weighted quadratic loss $\ell_\Gamma(z) = \frac{1}{2}(\Gamma z_-^2 + z_+^2)$, and the empirical plug-in estimator is shown consistent in Theorem 3. The paper's central practical warning is sharp. Allowing the Γ-bound to bite at every decision epoch ("per-decision confounding") yields bounds that grow exponentially in the horizon T, becoming uninformative for any realistic Γ when T is more than a handful of periods. The tractable regime is "one-decision confounding," where only a single nominated decision is potentially confounded. Their applied demonstrations on sepsis ICU management and a SMART trial for autistic children show the one-decision setting certifies policy contrasts up to design sensitivity Γ above five, while per-decision confounding loses informativeness already at Γ near 1.5.

Kallus and Zhou (2020) lift this analysis to the infinite-horizon setting and exchange per-step importance sampling for stationary-density ratios. Their Lemma 2 characterizes the partially identified set of state-occupancy ratios as solutions to an estimating equation involving the propensity ratio and an unobserved memoryless confounder, and their Theorem 1 establishes the sharpness of the resulting policy-value bound under a Tan-style marginal sensitivity model. The key structural assumption is that the unobserved confounder is conditionally independent of the prior history given the current state; with that, the bound problem reduces to a non-convex optimization that can be solved by alternating projections.

Bruns-Smith (2021) provide the practitioner-facing estimator that ties this sensitivity machinery back to the chapter's causal Bellman operator (Definition D2). Their Proposition 4 establishes a robust Bellman equation under the same memoryless-confounder assumption: with $\tau = \Lambda/(1+\Lambda)$ where Λ is the Tan-style odds-ratio bound,

$$\bar{Q}_t^\pi(s,a) \;=\; \mathbb{E}\Big[r_t(s,a,s') + \gamma \bar{V}_{t+1}^\pi(s') \,\Big|\, s,a\Big] \;-\; \alpha_t(s,a)\cdot \mathrm{CVaR}_{1-\tau}\Big(Y_t(\bar{Q}_{t+1}) \,\Big|\, s,a\Big). \quad (15)$$

The robust Q-function is the conditional expected shortfall of the Bellman target, and their orthogonalized loss yields a Neyman-orthogonal pseudo-outcome that relaxes the required nuisance rate from $o_p(n^{-1/2})$ to $o_p(n^{-1/4})$. Their MIMIC-III sepsis application picks a calibrated Λ between 1.4 and 2.5 and shows the robust policy shifts from fluid-heavy toward vasopressors, in line with prior clinical meta-analyses. The bridge to the existing chapter machinery is exact: Equation (15) is a Bellman operator with a CVaR penalty term that subtracts off the worst-case effect of an unobserved confounder under the Tan-style sensitivity bound.

For applied economics the practical hierarchy is the following. One claim is statistical: the logged data support a precise estimate under a maintained model. The other is identificational: the maintained model is strong enough to recover the interventional value. Sensitivity methods relax the second claim and report how quickly a policy recommendation changes as hidden bias grows. Namkoong et al. (2020) establishes that this can be done at all in the sequential setting and characterizes when it fails (per-decision confounding); Kallus and Zhou (2020) extends to infinite horizons; Bruns-Smith (2021) delivers a practitioner-ready estimator that plugs into standard fitted-Q pipelines. Recent conformal sensitivity work (Yin et al., 2024) shows the same instinct in static individualized treatment effects, although the RL setting is harder because per-period uncertainty is recursively priced through continuation values.

0.4.2 Proximal and POMDP Bridge Methods

When confounders are latent but proxy variables, noisy correlates of the confounder, are available, proximal causal inference provides identification. Bennett and Kallus (2023) adapt this logic to sequential settings by treating the problem as OPE in a POMDP. The analyst observes two proxies: a treatment-side proxy $W_t^{(1)}$ and an outcome-side proxy $W_t^{(2)}$, both conditionally independent given the latent confounder U_t. Identification proceeds through a bridge function h that solves a conditional moment equation linking the two proxies to the interventional quantity:

$$\mathbb{E}[V(S_{t+1}) \mid W_t^{(1)}, S_t, A_t] = \mathbb{E}[h(W_t^{(2)}, S_t, A_t) \mid W_t^{(1)}, S_t, A_t]. \tag{16}$$

The bridge function h is estimated by solving this integral equation, which reduces to a linear system in the discrete case, and the causal effect is recovered by marginalizing over the outcome proxy distribution.

$$P(s' \mid s, \mathrm{do}(a)) = \sum_{w_2} h(w_2, s, a)\, P(W^{(2)}{=}w_2 \mid s). \tag{17}$$

Two bridge functions, analogous to inverse propensity scores and Q-functions, yield a doubly robust estimator of the policy value that is $\sqrt{n}$-consistent without ever observing U_t. Bennett and Kallus (2023) demonstrate this idea in a sepsis management simulator in which the physician observes a true diabetes status that is only partially recorded in the analyst's data. Prior clinical observations serve as treatment-side proxies, current clinical observations serve as outcome-side proxies, and the proximal estimator ranks evaluation policies correctly in between 82% and 100% of test cases where naive estimators fail.

The POMDP bridge literature generalizes this insight beyond tabular examples. Shi et al. (2022) formulate OPE in confounded POMDPs as a minimax learning problem over bridge functions. Uehara et al. (2023) introduce future-dependent value functions that use future proxies as inputs and history proxies as instruments. Hong et al. (2024) move from evaluation to learning by identifying history-dependent policy gradients in confounded POMDPs. Wang et al. (2025) show a related proximal route in which past human or AI actions themselves can carry information about latent decision-relevant states.

0.4.3 Instrumental Variables and Orthogonal Scores

When neither backdoor nor front-door variables are available, instrumental variables can identify causal effects. An instrument Z_t must satisfy relevance, meaning it shifts the action A_t, and an exclusion restriction, meaning its effect on S_{t+1} is channeled entirely through A_t. Liao et al. (2024) formalize this as a Confounded MDP with Instrumental Variables (CMDP-IV), where transitions take the form $S_{t+1} = F^*(S_t, A_t) + \epsilon_t$ with unobserved confounders ϵ_t affecting both the behavior policy and transitions. The transition function F^* is recovered from a conditional moment restriction:

$$\mathbb{E}[S_{t+1} - F^*(S_t, A_t) \mid Z_t, S_t] = 0. \tag{18}$$

For a binary action and binary instrument, the Wald estimator provides a closed-form solution. Let β denote the causal effect of promoting (action $a = 0$) on the transition probability.

$$\beta = \frac{P(s' \mid s, Z{=}1) - P(s' \mid s, Z{=}0)}{P(A{=}0 \mid s, Z{=}1) - P(A{=}0 \mid s, Z{=}0)}, \tag{19}$$

where the numerator is the reduced-form effect of the instrument on the transition and the denominator is the first-stage effect on treatment uptake. The interventional transition is then recovered from any instrument value z via $P(s' \mid s, \mathrm{do}(a{=}0)) = P(s' \mid s, Z{=}z) + \beta \cdot (1 - P(A{=}0 \mid s, Z{=}z))$. Their IV-aided Value Iteration algorithm applies this moment restriction at each state to estimate F^*, then runs standard value iteration on the estimated model.

Liao et al. (2024) illustrate with neonatal intensive care unit (NICU) assignment. A hospital must decide whether to admit each newborn to the NICU, and this decision is confounded by unobserved severity indicators that affect both admission and infant health. Differential travel time to a specialty care provider serves as an instrument. Relevance holds because travel time shifts referral probabilities. The exclusion restriction requires that travel time affect infant health only through admission, not through any other channel.

Orthogonal-score methods solve a different problem. They do not by themselves remove hidden confounding, but they make causal OPE statistically stable when the identifying assumptions reduce the value to nuisance functions that must be learned flexibly. Liao et al. (2021) estimate long-term average policy values for mobile health applications. Lewis and Syrgkanis (2021) extend double machine learning to dynamic treatment effects through sequential residualization and Neyman-orthogonal moments. Kallus and Uehara (2022) derive efficient double reinforcement learning estimators for Markovian OPE using q-functions and stationary density ratios. The common econometric message is that identification and estimation are separate: first specify why the policy value is identified, then use orthogonal scores to keep first-stage machine learning error from dominating the target estimate.

0.4.4 Mediator-Based Identification

The front-door criterion applies when A_t affects S_{t+1} only through an observed mediator M_t. Three conditions are required: M_t intercepts all directed paths from A_t to S_{t+1}, no unblocked backdoor path exists from A_t to M_t, and A_t blocks all backdoor paths from M_t to S_{t+1}. When satisfied (Pearl, 2009),

$$P(s' \mid s, \mathrm{do}(a)) = \sum_m P(m \mid a) \sum_{a'} P(s' \mid s, m, a')\, P(a' \mid s). \tag{20}$$

The first factor is unconfounded; the inner sum adjusts for confounding on the $M_t \to S_{t+1}$ link by averaging over the observational action distribution. This is the sequential analogue of mediation analysis.

Mediator-based RL is most attractive when the action changes an observable channel before it changes the long-run state. In mobile health, an app prompt may affect cortisol through supplement adherence. In retail pricing, a promotion may affect conversion through a marketing follow-up or browsing response. Shi et al. (2024) use mediator variables to identify and construct confidence intervals for policy values in confounded MDPs. Ge et al. (2023) develop an RL framework for dynamic mediation analysis, decomposing the long-run effect into direct and mediated components. The policy question shifts from "does the action work" to "through which sequential channel does the action work, and is that channel stable enough to transport across policies?"

0.5 Structural Counterfactual Off-Policy Evaluation

The point-identification strategies of the preceding subsections recover the target-policy value $V(\widetilde{\pi})$ from observational data under explicit graphical assumptions. A complementary route, originating in Buesing et al. (2019), operates one level deeper. Cast the partially observable Markov decision process as a structural causal model, infer a posterior over its exogenous noise from the observed trajectory, and roll out under the counterfactual policy with that posterior held fixed. The output is not just a value estimate but a counterfactual trajectory the agent would have produced, sample by sample.

The transition mechanism is $s_{t+1} = f^s(s_t, a_t, u_t^s)$ with exogenous noise u_t^s, the observation mechanism is $o_t = f^o(s_t, u_t^o)$, and the action mechanism is $a_t = f^\pi(h_t, u_t^a)$ given the history h_t. Given an observed trajectory $\hat{\tau}$, the counterfactual rule proceeds in three steps.

Algorithm 1 Counterfactual off-policy evaluation (Buesing et al., 2019)

1: *Abduction.* Infer the posterior $P(U \mid \hat{\tau})$ over the exogenous noise from the observed trajectory.
2: *Action.* Replace the behaviour-policy mechanism f^π with the target-policy mechanism $f^{\widetilde{\pi}}$ to form the modified SCM $\mathcal{M}_{\hat{\tau}}^{\mathrm{do}(\widetilde{\pi})}$ with the same posterior on U.
3: *Prediction.* Sample $U \sim P(U \mid \hat{\tau})$ and roll out the modified SCM to obtain counterfactual trajectories τ^{cf} and their returns.

Lemma 1 of Buesing et al. (2019) establishes that the marginal of the counterfactual distribution over $\hat{\tau}$ equals the interventional distribution, so the counterfactual estimator is unbiased under correct SCM specification. On the partially-observable grid-world experiment of the paper, the counterfactual estimator's evaluation error decreases monotonically with the amount of off-policy data, while the model-based estimator's error remains stuck at the bias of the dynamics model.

The categorical-action case raises an identification problem. Oberst and Sontag (2019) show that multiple structural causal models can be consistent with the same interventional distribution yet imply different counterfactual outcomes, and they resolve the ambiguity by assuming a Gumbel-max sampling mechanism that satisfies a counterfactual-stability condition.[5] The shared identifying assumption across this strand is that the structural causal model is correctly specified. The Lucas critique applies in its sharpest form when the agent's policy itself enters the data-generating process, and identification by instruments or exogeneity restrictions in the spirit of Chernozhukov et al. (2018) is the econometric remedy that composes naturally with the backdoor, front-door, proximal, and mediator-based identification strategies of the preceding subsections.

0.6 The Broader Causal RL Landscape

Two further directions sit outside the chapter's central scope but deserve naming. Pace et al. (2024) respond to the case where no observed control, instrument, mediator, or proxy is strong enough to point-identify the policy value with Delphic offline RL, which treats hidden confounding as nonidentifiable uncertainty rather than pretending it has been solved. Mesnard et al. (2021) extends the structural-counterfactual machinery of Section 0.5 from policy evaluation to per-trajectory counterfactual policy gradients, using constrained future-information conditioning to keep the gradient estimator unbiased. Transportability theory (Pearl and Bareinboim, 2014; Bareinboim and Pearl, 2016; Li et al., 2024) asks which mechanisms remain invariant across source and target environments, governing sim-to-real transfer and curriculum learning under environment shift.

[5] The Pearl monotonicity condition resolves the binary case. The Gumbel-max SCM is the natural extension to k-way categorical outcomes. Tang and Wiens (2023) combine SCM-based counterfactual outcomes with importance sampling to obtain variance reductions while preserving unbiasedness under correct human annotation, with a sepsis-simulator demonstration that delivers an RMSE of 0.013 against 0.113 for vanilla per-decision IS.

0.7 Simulation Study: Confounded Retail Pricing MDP

I construct a 5-state engagement funnel $\mathcal{S} = \{0, 1, 2, 3, 4\}$ with two actions (promote, hold price) and an absorbing conversion state at $s = 4$. A retailer manages customers through engagement stages, deciding whether to offer a promotional discount. The data-generating process embeds four distinct sources of causal variation, enabling simultaneous validation of four point-identification strategies from a single DGP.

Figure 2 displays the complete causal graph. Market conditions $Z_t \sim \text{Bernoulli}(0.5)$ are observed and affect both the latent confounder and transitions. Consumer sentiment U_t is an unobserved confounder strongly correlated with Z_t.[6] An independent cost shock $\text{IV}_t \sim \text{Bernoulli}(0.5)$ serves as an instrument. The behavioral pricing policy depends on both U_t and IV_t: $\mu(\text{promote} \mid s, U_t, \text{IV}_t) = 0.55 + \rho \cdot 0.25 \cdot (2U_t - 1) + 0.15 \cdot (\text{IV}_t - 0.5)$, where $\rho \in \{0, 0.2, \ldots, 1.0\}$ controls confounding strength. Promotions trigger marketing follow-ups with M_t serving as a mediator.[7] Two noisy proxies of U_t are available: a CRM score $W_t^{(1)}$ and browsing behavior $W_t^{(2)}$.[8]

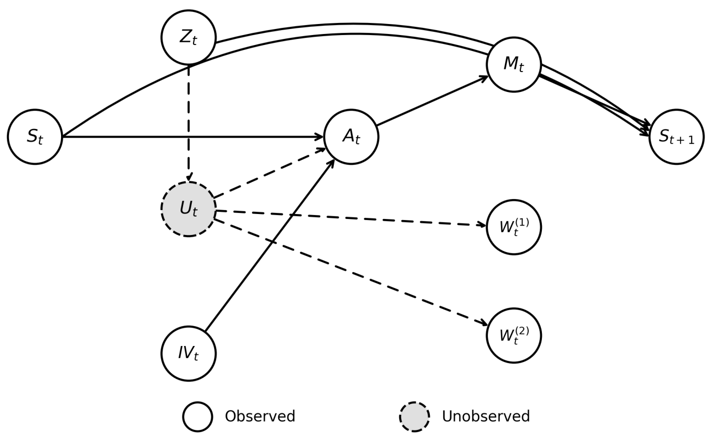

Figure 2: Complete causal graph of the simulation DGP. Gray dashed node (U_t) is unobserved; dashed edges involve U_t.

The action affects the next state only through the mediator: $P(s{+}1 \mid s, M_t, Z_t)$ de-

[6] $P(U_t{=}1 \mid Z_t{=}1) = 0.9$ and $P(U_t{=}1 \mid Z_t{=}0) = 0.1$.
[7] $M_t \sim \text{Bernoulli}(0.8)$ when the retailer promotes and $\text{Bernoulli}(0.2)$ otherwise.
[8] $W_t^{(1)} \sim \text{Bernoulli}(0.85 \cdot U_t + 0.15 \cdot (1 - U_t))$ and $W_t^{(2)} \sim \text{Bernoulli}(0.75 \cdot U_t + 0.25 \cdot (1 - U_t))$.

pends on M_t and Z_t but not on A_t or U_t directly. This enables all four identification strategies simultaneously: Z_t satisfies the backdoor criterion, M_t satisfies the front-door criterion, IV_t satisfies relevance and exclusion, and $W_t^{(1)}, W_t^{(2)}$ satisfy the proximal conditions.[9]

I compare six estimators: oracle, naive (biased per Lemma L1), backdoor (Equation 13), front-door (Equation 20), Wald IV (Equation 19), and proximal (Equations 16 and 17).[10] These estimators correspond to the point-identification rows of Table 1. The simulation intentionally leaves out sensitivity bounds and orthogonal-score refinements so that the figure isolates the identification logic from higher-order statistical efficiency.

Figure 3 reports bias and RMSE across confounding strengths (20 seeds, 2,000 trajectories per seed). The naive estimator's bias grows monotonically with ρ because observational transitions overestimate promotion success: the promote action is more likely when $U_t = 1$, which correlates with favorable conditions, so $\hat{P}_{\mathrm{obs}}$ exceeds the true interventional probability, and this per-step bias compounds through the Bellman recursion. The backdoor and front-door estimators eliminate bias at all ρ, validating Theorem 1 and Equation (20). The IV estimator maintains low bias but higher variance due to the Wald ratio's sensitivity to instrument strength; panel (c) illustrates this classic bias-variance tradeoff. The proximal estimator achieves low bias with moderate variance, confirming that bridge functions recover causal effects from noisy proxies.

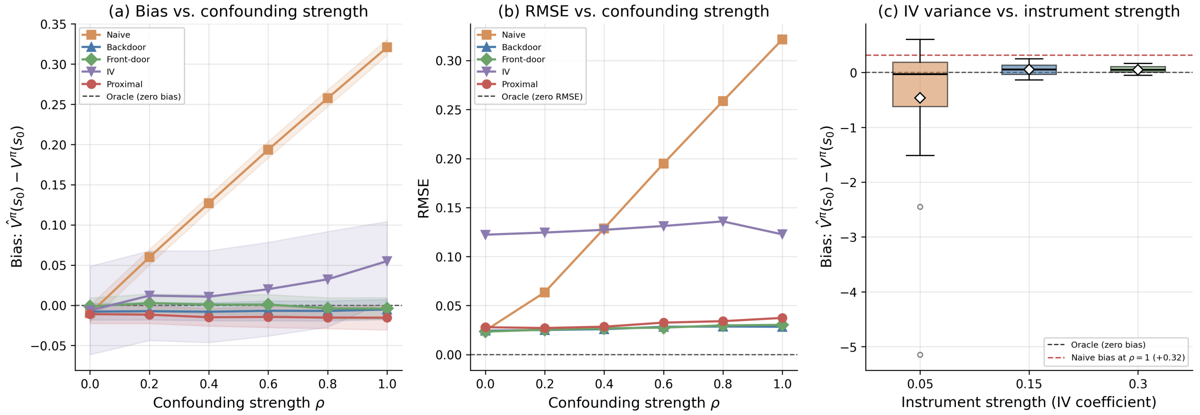

Figure 3: (a) Bias of five OPE estimators as a function of confounding strength ρ. (b) RMSE of five estimators as a function of ρ. (c) IV estimator bias distribution vs. instrument strength at $\rho = 1$; dashed red line is naive estimator bias.

[9] Rewards are $r(s, a) = -1$ for $s < 4$, $\gamma = 0.9$. The target policy always promotes. The true interventional transition probability is $P(s{+}1 \mid s, \mathrm{do}(\mathrm{promote})) = 0.615$.

[10] Each configuration uses 2,000 trajectories averaged over 20 seeds.

0.8 Simulation Study: Counterfactual OPE under a Misspecified SCM

A second synthetic instance illustrates the counterfactual machinery of Section 0.5 on a contextual-bandit substrate that strips the temporal complication and isolates the model-misspecification question. The DGP is a linear structural causal model in two features,

$$y = 1 + 0.5\,x_1 - 0.3\,x_2 + 2\,a + a \cdot x_1 + u, \qquad u \sim \mathcal{N}(0,\, 0.5^2), \tag{21}$$

with $x \sim \mathcal{N}(0, I_2)$, a binary action $a \in \{0, 1\}$, and a heterogeneous treatment effect $\tau(x) = 2 + x_1$. The behaviour policy $\pi_{\text{obs}}(a = 1 \mid x) = \sigma(0.5 + x_1 - 0.5\,x_2)$ is confounded with x. The target policy $\tilde{\pi}(a \mid x) = \mathbf{1}\{x_1 + x_2 > 0\}$ is a deterministic threshold rule. The oracle value $V(\tilde{\pi}) = \mathbb{E}_x[y \mid a = \tilde{\pi}(x)]$ is computed by Monte Carlo at 10^6 realisations.

Three estimators are compared on $n \in \{200, 500, 1000, 2000\}$ logged samples, 20 seeds per cell. Per-decision importance sampling is $\widehat{V}_{\text{IS}} = n^{-1} \sum_i \rho_i y_i$ with $\rho_i = \mathbf{1}\{a_i = \tilde{\pi}(x_i)\}/\pi_{\text{obs}}(a_i \mid x_i)$. The model-based plug-in $\widehat{V}_{\text{MB}} = n^{-1} \sum_i \hat{f}(x_i, \tilde{\pi}(x_i))$ uses an OLS fit $\hat{f}$ on the logged data. The counterfactual estimator follows the abduction-action-prediction rule of Algorithm 1 and combines $\hat{f}$ with a residual correction weighted by the importance ratio,

$$\widehat{V}_{\text{CF}} = n^{-1} \sum_i \hat{f}(x_i, \tilde{\pi}(x_i)) + n^{-1} \sum_i \rho_i \Big(y_i - \hat{f}(x_i, a_i) \Big), \tag{22}$$

which is the doubly-robust form of the abduction-action-prediction rule and is unbiased under correct propensity *or* correct outcome model.[11] The well-specified scenario uses the feature set $\{1, x_1, x_2, a, a \cdot x_1\}$ in $\hat{f}$, matching the DGP. The misspecified scenario drops the interaction $a \cdot x_1$, the source of the heterogeneous treatment effect.

Table 2 reports bias, standard deviation, and root mean squared error at $n = 1000$. Under well-specification the model-based estimator attains bias -0.001 and RMSE 0.056, the counterfactual estimator attains bias 0.003 and RMSE 0.065, and the importance-sampling estimator attains bias 0.017 and RMSE 0.099. The MB and CF estimators are statistically indistinguishable and both improve on IS. Under misspecification the MB estimator's bias jumps to -0.072 and its RMSE to 0.093, while the CF estimator's bias

[11] The naive average of the per-observation counterfactual outcome $y_i' = \hat{f}(x_i, \tilde{\pi}(x_i)) + (y_i - \hat{f}(x_i, a_i))$ collapses to $\widehat{V}_{\text{MB}}$ under OLS with an intercept because the OLS residuals sum exactly to zero. The importance-weighted residual correction in (22) restores the bias-cancellation property in finite samples. The estimator coincides with the classical doubly-robust estimator of Robins-Rotnitzky-Zhao adapted to off-policy evaluation.

Table 2: Counterfactual off-policy evaluation under a linear SCM, $n = 1000$, 20 seeds. The CF estimator's residual correction cancels the MB bias under misspecification while inheriting low variance from the model.

Scenario	Estimator	Bias	Std	RMSE
Well-specified	IS	$+0.017$	0.100	0.099
	MB	-0.001	0.057	0.056
	CF	$+0.003$	0.067	0.065
Misspecified	IS	$+0.017$	0.100	0.099
	MB	-0.072	0.061	0.093
	CF	$+0.002$	0.078	0.076

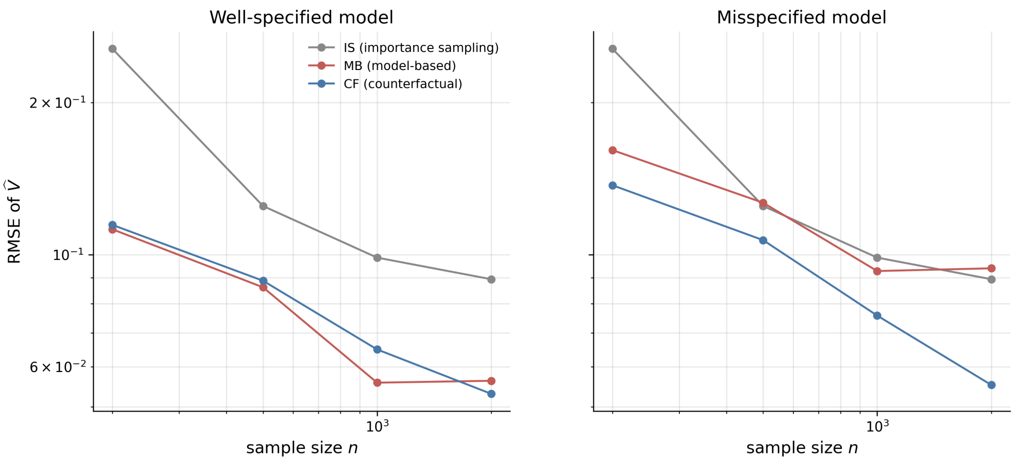

Figure 4: Root mean squared error of three OPE estimators against the sample size, log-log scale, two panels. Left panel, well-specified outcome model. Right panel, misspecified model with the interaction omitted. The MB estimator's RMSE plateaus under misspecification as the bias dominates the variance, while the CF estimator's RMSE continues to decline.

remains at 0.002 and its RMSE at 0.076. The CF estimator's RMSE is 18% lower than MB's, and the bias is two orders of magnitude smaller. The IS estimator is unchanged across scenarios because it does not use an outcome model. Figure 4 plots RMSE against n on log-log scale. The MB curve flattens under misspecification at the population bias of the OLS projection, the CF curve continues to decline at the parametric rate, and the IS curve traces the higher-variance parametric rate that the importance-weighted estimator delivers without any model assistance.

The companion Chapter **??** inverts the direction of this one, asking how RL machinery imports into econometric causal inference rather than the reverse. The bridge there runs through the equivalence between Murphy's backward Bellman recursion for optimal dynamic treatment regimes and Watkins's Q-learning recursion on a history-augmented state. Once that translation is fixed, the dynamic-DML and policy-learning results of that chapter complement the identification techniques of this one: this chapter deals with what to do when sequential ignorability fails; the companion chapter deals with how far you can push the rate of convergence when it holds.

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